



SAFETY Department
SIMHADRI

Note No :1



Ref:SMTTPP/SAFETY/2023-24/LMI/SAFT/643200

Date:09/11/2023(18:06:15)

Subject:Approval of LMI on Job Safety Analyses & Method Statements -reg.

Sub:- Approval of LMI on Job Safety Analyses & Method Statements -reg.

The LMI on Job Safety Analyses & Method Statements, Ref no: LMI-OD-OPS-SAFT-003, Dated: NOV - 2023 is reviewed and no changes incorporated in the LMI.

This Supersedes LMI-OGN-OPS-SAFT-003 , Dated: AUG -2021.

Put up for approval of Competent authority.

Encl:-

1. New LMI (LMI-OD-OPS-SAFT-003, Dated: NOV -2023)
2. OD (COS-ISO-00-OD/OPS/SAFT/003 Rev. No. 0 June-2021)
3. Old LMI (LMI-OGN-OPS-SAFT-003 , Dated: AUG -2021)

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Note No:2

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Note No:11

Few gramatical errors observed. Same changed in word format and highlighted.

Forwarded for further review and approval.

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Reviewed.

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May please review

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Note No:25

JSA format in local Vernacular Language (TELUGU) may also be provided.

Reviewed and forwarded.

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SIMHADRI

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

Local Management Instruction – LMI/OD/SAFT-003

(COS-ISO-00-OD/OPS/SAFT/003 Rev. No. 0 June-2021)

Revision: 00 Dt: NOV 2023

Job Safety Analyses (JSA)
&
Method Statements (MS)

Approved for Implementation by.....
HEAD OF PROJECT (Simhadri)

Date:..... 11.2023

SAFETY DEPARTMENT

TITLE OF LMI:		<i>Job Safety Analyses (JSA) & Method Statements (MS) of activities at Simhadri Super Thermal Power Station.</i>					
LMI No:		LMI – OPS – SAFT- 003					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
01	00	.08.21	All clauses are new as the document is being issued for the first time.	P S S REDDY	ALL HODs	GM (O&M)	HOP
02	00	.11.23	NIL	Vigneshwar D	ALL HODs	GM (O&M)	HOP

N T P C L I M I T E D

SIMHADRI SUPER THERMAL POWER STATION

LMI APPROVAL DOCUMENT

Title of LMI:	<i>Job Safety Analyses (JSA) & Method Statements (MS) of activities at Simhadri Super Thermal Power Station.</i>	
LMI No.:	LMI –OPS–SAFT-003	
Issue No. : 2	Revision: 0	Date : .11.2023

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Date of Issue			

Job Safety Analyses (JSA) & Method Statements (MS)

INDEX

	Page No
1.0 Introduction	1
2.0 Superseded Documents	1
3.0 Revision Details	1
4.0 Scope	1
5.0 Roles & Accountability	2
6.0 Requirements.....	3
6.1 Appointment of Assessors	3
6.2 System of Written Safe Work Procedures.....	3
6.3 Method Statements	3
6.4 Job Safety Analysis	6
7.0 Records.....	8
8.0 Audit.....	9
9.0 Review	9
Annexure 1 - Information Expected To Be Included In Method Statements.....	10
Annexure 2 - Typical JSA – Job Safety Analysis	14

Job Safety Analyses (JSA) & Method Statements (MS)

1.0 Introduction

NTPC recognizes in the Safety Policy that *all accidents are preventable*. Under the NTPC Safety Framework, the basic requirement is to examine what could cause harm to people as a result of work activities and to make a judgment about whether enough precautions have been taken or if further actions are required. The process for carrying out such Hazard Identification & Risk Assessments for high risk processes, have been issued in the form of Operations Directives.

This directive covers the related aspects of work methods assessment to prevent incidents – Method Statements (MS) and Job Safety Analysis (JSA). This OD does not contain elements that can vary as per location. Hence no modification or, adaptation for location is expected for this OD.

2.0 Superseded Documents

Nil

3.0 Revision Details

Nil

4.0 Scope

The application of the principles described in this guidance in construction projects will assist NTPC's Area Engineers to assess the adequacy of a Method Statement, to formally review its contents and to meet the requirements of contract management in respect of Supervision. And, in operational stations, to serve as information to Senior Authorized Persons for ensuring effective safety from the system. The guidelines also serve in projects when NTPC has been engaged as a consultant by a client and is expected to produce a Method Statement.

Job Safety Analysis (JSA) & Method Statements (MS) are similar in intent but vary in detail.

JSA is a written procedure developed to identify the dangers of a particular task in order to reduce the risk and complete the work safely without any incident.

A Method Statement is much more detailed and aims to provide adequate levels of information across all elements of the work in relation to health and safety having particular regard to any associated hazards / risks. Method Statement will consider 'what', 'when', 'where' and 'how' of the work to be done. Foreseeable hazards should be identified and risk assessments provided to demonstrate

suitable control measures. The detail required must be proportionate to the level of risk.

Both JSA and MS can only be prepared once the potential hazards have been identified and the risks assessed. This therefore requires a properly considered risk assessment for each activity, or series of activities, which identifies not only the hazards, but also the required control measures that are necessary to manage the residual risks. Thus, a Risk Assessment is embedded in both JSA, in a brief manner and in much more detail in a MS.

5.0 Roles & Accountability

- Head of the Project/Stations (HOP) is accountable for ensuring that these directives are implemented without modification at their location; to ensure in place a system to manage high-risk tasks effectively through formal written safe work procedures (SWP) that necessarily include Job Safety Analyses or, Method Statements.
- In construction projects, Head (Construction) shall be responsible: to implement this procedure; to ensure availability of adequate safe work procedures from contractors and; to ensure that all Area Engineers controlling contractors are trained in this procedure.
- In operational plants, Head (O&M) shall be responsible: to implement this procedure; to ensure safe work procedures are implemented and; that Area Engineers and SAPs are trained in this procedure.
- For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for the above roles.
- Head (Maintenance) shall ensure safe work procedures are implemented, specific Job Plans are developed for the location, along with specific Job Safety Analyses or, where applicable specific Method Statements.
- Area Engineer shall be responsible for ensuring that the SAP (Senior Authorized Person) is in receipt of all necessary relevant information on the planned work to enable a correct assessment of the Safety Document requirements to be made.
- Senior Authorized Person (SAP) shall ensure that all necessary safety precautions are carried out to establish "Safety from the System" and indicate any further precautions to be taken during the course of the work to avoid system derived hazards, in accordance with NTPC Safety Rules (Electrical and Mechanical).

- Site Safety Coordinator is accountable to ensure that Site safety assessments cover availability, adequacy and implementation of MS/JSAs and to ensure findings of such assessments as discussed in periodic Safety Committee Meetings and included in periodic functional reports.

6.0 Requirements

6.1 Appointment of Assessors

Each HOS will nominate persons from within their department to undertake the duties of a risk assessor. Requirements of OD on Hazard Identification & Risk Assessment shall be met with in the process of Appointment of such Assessors.

Responsibility:- All Head of Sections

6.2 System of Written Safe Work Procedures

All projects shall have in place written Safe Work Procedures (SWPs) in the form of Standard Operating Procedures (SOPs), Standard Maintenance Procedures (SMPs), Job Plans (or Work Instructions (WI)) for managing high- risk tasks, as identified through risk assessments. Such SWPs should consider different literacy levels. The SWPs are to take cognizance of identified hazards and risks, work steps/sequence, tools, equipment required, and actions in case of emergency (shutdown, spill handling, isolation, etc.). Protective equipment/clothing required is to be indicated. Critical health & safety related steps/controls are to be identified/highlighted and employees/workers made familiar therewith.

All works in the project shall have as a minimum, a JSA attached to the Work Request (physically or, electronically). Where electronically attached, the JSA/MS should be a secured document like a PDF file where the contents cannot be changed. For high risk or, larger scope works, job specific Method Statements are necessary.

Responsibility:- All Head of Sections

6.3 Method Statements

6.3.1 Method Statement - Principles

A Method Statement is a document containing information about work. The aim is to provide adequate levels of information across all elements of the work in relation to health and safety having particular regard to any associated hazards / risks. Foreseeable hazards should be identified and risk assessments provided to demonstrate suitable control measures. The detail required should not be excessive but must be proportionate to the level of risk. Information provided should be relevant or necessary and be adequate. Care should be taken such that unnecessary or irrelevant information is not included as in such cases important omissions may be masked by the mass of irrelevant data provided. Job specific Method Statements may be necessary where hazards warrant them as in the case of high risk tasks.

Much of the work carried out on NTPC construction sites will require the submission of a "Method Statement" esp., those involving heavy lift, excavation, trenching, work at height.

In operational projects, Method statements are expected for hazardous tasks/works such as or involving: work at height, hot work, confined spaces, heavy lifting, work with hazardous chemicals and working in hot environments.

When the proposed works are minor in nature, a single Method Statement may be sufficient. For large or, complex projects, as in construction phase, however, a series of Method Statements should be in place, each of which would cover individual elements of the total work program. When amalgamated they would comprise a major part of the overall 'Safety Plan' which will consider 'what', 'when', 'where' and 'how' the work will be done.

Responsibility: Head of Sections

6.3.2 ASSESSMENT OF A METHOD STATEMENT

In Construction phase, there would be a number of times, where contractors submit Method Statements (MS) for high risk works. In Operational phase, MS are essential for high risk tasks like work at height and non-routine works like overhauls. For Operational projects, these guidelines also serve as a means to ensure that the Method Statements submitted by the Area Engineers cover all the essential aspects of their proposed work.

When assessing the adequacy of a Method Statement, or for deciding whether a Method Statement is necessary at all for work where hazards / risks are insignificant, the following points should be considered. It must be emphasized that the detail required is proportionate to the foreseeable risk and while consideration should be given to each of the points below it may well be the case that information will not be required if the level of hazard / risk does not warrant it.

- Detail of engineering requirements, what work is to be done and how the contractor proposes to carry it out. Any boundaries, limitations or restrictions should be included.
- Detail of 'significant' tools, equipment, machinery, materials, processes, substances or PPE to be used during the course of work.
- Detail of any requirements for access, transport, mechanical handling.
- Identification of hazardous elements of the work, evaluation of risks from risk assessments detailing appropriate control measures (including appropriate drawings, diagrams or plans).
- Details of the levels of supervision of staff, involved with hazardous activities.

- Confirmation of compliance with applicable legal requirements ex. BOCW Act, OSH Code etc.
- Details of how contractors' employees or those of any sub-contractor will be informed of the relevant control measures appropriate to the work they are undertaking – like Tool Box talks, training.
- Details of contingency plans for dealing with foreseeable emergency situations like safe evacuation from height.
- Estimates of the time scales for the work program particularly in relation to the duration of any hazardous activities.
- In deciding whether the information provided is sufficient, ensure that vague statements such as "hazards will be identified" or "emergency procedures will be established" are not accepted.
- The Method Statement should indicate how contractors' direct employees or those of any sub-contractor will be informed of the relevant control measures appropriate to the work they are undertaking.

Annexure – 1 further enumerates non-exhaustive guidelines for assessing Method Statements and includes a typical Method Statement template and typical training record form.

Responsibility: Head of Sections

6.3.3 'ADEQUACY' OF METHOD STATEMENTS

In general, adequacy of the Method Statements (MS) is ensured by relevant NTPC employees (Area Engineers) who review the MS and to make an assessment as to whether or, not the work can proceed in a safe manner in accordance with NTPC procedures. Where NTPC engages contractors for their specialist knowledge and expertise, it may be possible that the location staff will be unable to make a formal assessment of those elements of the work that are outside the parameters of their own knowledge and experience but, will note the intentions as laid down by the MS. For all cases, the Area Engineers should be capable of questioning the contractor to establish that the proposed works can proceed safely. If the Area Engineer does not have the technical competence to assess the hazards from the MS, another competent person should be assigned that work by the HOS. Work should not be allowed to start unless the way it is going to be carried out is understood by Area Engineer/HOS.

Responsibility: Head of Sections

6.3.4 IMPLEMENTATION OF METHOD STATEMENT

In construction projects, direct supervision of contractors' employees is the responsibility of the contractor. NTPC Area Engineer should not provide direct supervision to contractors' staff but must exercise sufficient levels of managerial control. This may sometimes involve a presence at the point of work but, will normally be directed towards ensuring that the contractor is providing the

necessary levels of supervision for his employees to take account of the foreseeable hazards / risks as previously identified in the Method Statement.

In operational projects, the NTPC Area Engineer shall be responsible to ensure the necessary levels of supervision of workers / outsourced contractors to take account of the foreseeable hazards / risks as previously identified in the Method Statement.

Responsibility: Head of Sections

6.3.5 MONITORING OF IMPLEMENTATION OF METHOD STATEMENT

Site Safety Coordinator is accountable to ensure that site safety assessments cover adequacy and implementation of MS/JSAs. Site Safety Coordinator along with Area Engineer / respective HOS shall make periodic ground level assessments of implementation of the MS. Deviations observed during field assessments should be analyzed to establish whether safe work procedures need to be amended or, the employee/contractor needs additional training.

Site Safety Coordinator shall ensure findings of such assessments are captured in the centralized system; trends should be analyzed and the associated system weaknesses identified; discussed in periodic Safety Committee Meetings; action planned to rectify system weaknesses, and; included in periodic functional reports.

Responsibility: Site safety Coordinator

6.4 Job Safety Analysis

6.4.1 JSA Principles

A Method Statement in a simplified form is sufficient for routine work in the operational plants. In this simplified form the MS is termed as JSA. A JSA is thus a task-oriented risk assessment used to review the hazards associated with a particular job plan, and to verify that adequate safeguards are in place to control those hazards. The main goal of this analysis is to prevent harm to the individuals carrying out the task. JSAs are known by a variety of terms, including Job Hazard Analysis (JHA), Job Risk Assessment (JRA), Task Risk Assessment (TRA).

The NTPC procedure of requiring a JSA to be performed for jobs requiring a PTW is a safety best practice aimed at achieving further risk-reduction. An effective JSA identifies and controls the risks that are specific to the current location and current conditions, such as simultaneous work at different levels on a boiler, which a PTW alone may not have considered.

6.4.2 What-If Analysis

Incidents commonly result from deviations from work practices or, from deviations in the expected operation of tools/tackles/equipment (failures). Thus, JSAs should consider a “What-If” analysis, of what could go wrong in each

particular job step, as a technique for hazard identification. A “What-If” analysis will consider failures and deviations that are reasonably possible, and omit the highly unlikely ones.

To illustrate this approach, consider the example of a routine using the hoist for lifting a gas cylinder in position. The JSA indicates a step for hoisting the gas cylinder and should consider the reasonable What-If possibilities like "What-If the hoist is stuck mid-way?"; while the possibility of hook failure may be discounted as unlikely possibility as the hoist would have been tested for full load as per statutory requirements. In this case, for the unlikely hook failure, restricting access to the area below the location where the gas cylinder is being lifted is a reasonable control measure.

6.4.3 JSA Process

The Key Steps in JSA are summarized below:

1. Input for the JSA is the specific Job Plan for the work, and for each sub-task of the Job Plan:
2. Identify potential hazards for each sub-task of the Job Plan;
3. Identify available / planned risk control measures/precautions;
4. Assess the residual risk (i.e. after control measures);
5. Identify who would be responsible for implementing the control measure;
6. Identify what protective and safety equipment should be used.

The Area Engineer shall ensure that assessed risks are addressed, and that decisions are clearly documented.

7. Determine if work can proceed. If risk identified is high, stop the activity, carry out detailed HIRA as per the OD on HIRA, and prepare a Method Statement, as per guidelines provided elsewhere in this document.

8. Ensure all persons involved in the work understand the relevant hazards & risks before work starts.

6.4.4 Library of Generic JSAs

Projects may decide to keep a library of pre-developed JSAs for reference by the location assessors in the development of future JSAs. Generic JSAs refer to JSAs performed for common tasks such as “replacement of motor”, covering basic steps and basic hazards expected. They are general enough to be applied across the project.

Responsibility: Head of Sections

While the main advantage in the use of generic JSAs is achieving consistency in identifying the subtasks, hazards, and controls for each generic task. Pre-filled JSAs have the danger of being used blindly by the personnel without truly performing a risk assessment for the current conditions. Another potential downside of the use of completed JSAs is that task personnel may not assume ownership of the JSA process.

Hence, generic JSAs should never be used directly as the final JSA for the task. There are many hazards that are unaccounted in generic JSAs that must be addressed at the worksite, (e.g., the current weather, the time of day, the experience of the task personnel, the condition of the tools used in the job, the location of the worksite, whether there are proximate conflicting operations etc.).

Responsibility: Head of Sections

JSA template is included as Annexure – 2

7.0 Records

Any record that is maintained as an electronic copy shall be in a format that cannot be edited. The following records shall be maintained:

All HOSs shall maintain for their respective functions:

- JSA/MS register: the table of equipment that contains the list of equipment (or areas) with specific job plan number (or work instruction) and list of specific JSAs against each job plan/work instruction;
- Evidence of a system for implementation of the specified control measures as per JSA/MS in the form of: a table laying out: works for the day, reference number of the JSA/MS for that work, person accountable for implementing risk controls specified in the JSA/MS, person accountable for delivering tool box talk on the hazards of that work, person accountable for supervision of the work.
- Evidence of a system for checking implementation of the specified control measures as a Department of Functional schedule of reviews/inspections.
- Evidence of tool box talks.
- Evidence of corrective action (including exemplary penal action) taken when such specified control measures were not implemented or violated.

In case of construction projects: All HOSs shall maintain a table giving the number of JSA/MS available against number of job plans/critical erection/project activities. Such table should include accountable role for supervision.

Completed MS and JSAs of high risk tasks should be used for operator training, incident investigation, hazardous substance review, etc., and should be kept for future reference for a period of 3 years as scanned electronic copies.

8.0 Audit

Evaluation of the compliance with this OD shall be carried out at least once annually. The compliance evaluation assessment/audit can be included as part of other compliance audits or assessments like technical compliance audits.

The audit shall be a combination of verification of evidence in the form of records, verification of implementation in the field and interviews.

Records: Audit shall include verification of all the records detailed under 7.0.

Implementation: Auditors shall validate the evidence of a system for verifying control measures by the function/department/section and verify implementation of such system. evidence of action initiated, including exemplary penal action, for failure to implement risk control measures shall be verified and included in the audit report.

Field audit shall be in the form of random assessments. Auditors shall take a random sample of at least five works/activities from at least three different functions, including one non-technical function to assess the implementation of the committed control measures including supervision and tool box talk requirements.

Interviews: check with at least five workers randomly selected from different functions on their awareness of hazards of the task they are carrying out; HOS of respective function shall be accountable for demonstrating compliance to implementation and awareness requirements. Activities verified and persons interviewed shall be part of the audit report.

For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for ensuring periodic audits of the system at such projects under his/her control and maintain records.

9.0 Review

The Head of Project shall arrange for review of this LMI by Head of Safety as and when required, or every two years.

Annexure 1 Information Expected To Be Included In Method Statements

The content of the Method Statement will depend on the complexity and size of the job. Its purpose is to show how the work will be carried out safely. Method Statements should address each of the following, where applicable:

- Details of the organization in control of the activity.
- Details of the supervisor / manager responsible for the activity with contact number.
- Details of Site contacts including safety officer, emergency numbers etc.
- Document number, issue date, revision date, revision number etc.
- Start date, completion date
- A brief description of the activity.
- Written work Hazard Identification & Risk Assessment.
- Work permits such as the Permit to Work and any other legal requirements.
- Work location, its boundaries, access & how it is segregated from other activities.
- Equipment to be used – list of plant/equipment/tools/tackles to be used in the execution of the work, its safe working load, details of test certificates, inspections, registers.
- A way to prevent unauthorized access to the work area and beneath the work area if work is at height, where the work is being undertaken
- A step by step description of the activities to be undertaken.
- Methods of executing the work, including methods of lifting, fixing, holding or bolting.
- Methods to prevent any falls from heights incl., details such as working platforms, handrails, safety harnesses, safety nets, lifelines or, other means of preventing falling.
- Access and egress to the job, ex. by ladders, mobile work access platforms, hoists etc. including the location of any ladders or other means of access.
- Methods of protecting materials falling from heights ex. toe boards, debris guards, platforms.
- When required, detailed calculations for any loading of platforms, props, temporary works or supports that are to be provided during the progress of the job.
- Details of any confined space hazards and where necessary, atmospheric monitoring procedures and emergency equipment to be provided.
- Stabilization of the works during their progress ex., any temporary props, struts or supports that are required.
- Details of any shoring to be provided in excavations, means of entry and barriers or secure coverings to be provided.
- Precautions necessary to protect workers, and other people that could be affected, including personal protective equipment.
- Training procedures, tool box talks, specific training on risks etc.

- Emergency procedures, including the location of emergency equipment.
- The procedure for changing the proposed method of work if necessary.
- Details of storage and the stacking of items together with any special delivery procedures and any assembly work that is going to be carried out.
- When applicable, the method for safeguarding existing structures.
- Details of grade of workers involved ex., fitters, riggers, crane operators, helpers etc. If there is any need for specially-trained operators for certain activities.
- The training of operatives on site ex., induction training and any details regarding their part to be played within the Work Method Statement

Annexure 1 – FORMAT FOR METHOD STATEMENT

Description of the Task/Activity:			
Project Name:		Project Ref:	
Start Date/Time:		Finish Date/Time:	
Personnel involved:	Name	Role/Trade	
Work Supervisor:		Phone:	
Key Plant and Tools Required:			
Key Materials Required:			
Other Essential Equipment:			
Specific Identified Residual Hazards: (or refer to the task specific risk assessment(s))			
Specific Training Requirements:			
Sequence of Operations: (Specifying methods of working, tools, materials and equipment utilized)			
Temporary Supports and Props needed to facilitate the works:			
Method of Access and Egress to the work area:			
Fall Protection Measures: (consider both Personnel & Materials)			

COS-ISO-00-OD/OPS/SAFT/003 Rev. No.: 0
Job Safety Analyses (JSA) & Method Statements (MS)

Hazardous Chemicals / substances involved? If YES, specify the details, attach MSDS	
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Required Personal Protective Equipment	
Emergency Procedures	
First Aid contact	
Any Other Information	

All work will be undertaken by qualified competent persons with experience of the type of work described above, and in all cases in full accordance with safety procedures specified in the company's Health and Safety Policy.

Prepared by: _____ **Reviewed by:** _____ **Date:** _____

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TYPICAL METHOD STATEMENT/JSA BRIEFING RECORD - TRAINING/TOOL BOX TALK

Training delivered by: _____ *Dept.:* _____ *Date:* _____

We (the undersigned) have read and understood the attached method statement and will comply with the specified requirements and control measures. If the work activity changes or deviates from that originally envisaged, we will seek further advice and request an amended method statement.

Name (Print)	Signature	Date

Annexure 2 – Format for JSA – Job Safety Analysis

JOB SAFETY ANALYSIS	JSA Number:		Date:	
	JSA DONE BY:			
Work Instruction#/ JOB PLAN #			PTW/LWC #	

JOB (Brief description)					
JOB LOCATION				DEPARTMENT	
#	JOB PLAN STEPS (add additional rows as required)	HAZARDS	CONTROL MEASURES	RESIDUAL RISK (High/Medium/Low)	PERSON RESPONSIBLE FOR CONTROL MEASURE

PPE REQUIRE D (Yes/No)	Helmet	Safety Shoe s	Gloves	Eye/Face Protectionn	Hearing Protectionn	Respiratory Protection	Other PPE	Other PPE
ADDITIONAL PRECAUTIONS / CONTROL MEASURES, IF ANY:								
FURTHER PRECAUTIONS / CONTROL MEASURES, IF ANY:								
TOOL BOX TALK /								

COS-ISO-00-OD/OPS/SAFT/003 Rev. No.: 0
Job Safety Analyses (JSA) & Method Statements (MS)

TRAINING TO BE GIVEN BY:			
This is to certify that all work under the scope of the above Job Plan shall be undertaken by qualified competent persons with experience of the type of work described above, and in all cases in full accordance with NTPC safety rules (E&M) and conditions of the safety document. What-If failure scenarios have been considered like failure of tools, tackles, work equipment in this JSA.			
	Area Engineer Name		HOS* Name
	Signature		Signature

*HOS = Head of Section